

Reduction of PCB Contamination in an Arctic Coastal Environment: A First Step in Assessing Ecosystem Recovery after the Removal of a Point Source

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Studies have documented the presence of organic contaminants in Arctic marine environments due to local point sources, but little long-term data is known of the recovery process once those contaminants are removed. In this study, natural recovery of a local marine food web from a historic terrestrial source of polychlorinated biphenyls (PCBs) at Saglek Bay is examined. PCBs were measured in marine sediments, shorthorn sculpin (*Myoxocephalus scorpius*), and black guillemots (*Cepphus grylle*) to assess ecosystem recovery over an eight-year period after cleanup of the source. Surface sediment measurements indicate decreases in both the extent and average concentrations of PCBs throughout Saglek Bay. Sculpin collected near the former contaminated beach in 2006 had significantly lower PCB burdens than those collected in 1998/99. PCB levels have also decreased significantly in black guillemot nestlings collected from the former contaminated beach area, as well as Island (6–8 km from the source) and Reference (greater than 18 km from the source) locations. This study demonstrates the utility of using indicator species, especially the black guillemot, to monitor PCB uptake in benthic food webs. The rapid decreases in the physical and biotic PCB concentrations from the local ecosystem at Saglek Bay demonstrate the efficiency with which natural ecosystem recovery can take place in a highly energetic coastal marine environment once a chronic contaminant source is removed.

Introduction

Polychlorinated biphenyls (PCBs) are present in the Arctic as a consequence of both long-range atmospheric transport and local point sources, such as military sites (1–4). Saglek Bay, Labrador, has been the site of a military radar station (LAB-2) since the late 1950s (5, 6). During an environmental investigation of LAB-2 in 1996, extensive PCB contamination was found in soil at the site, and subsequent work found

elevated PCB concentrations in local sediments, invertebrates, fish, and seabirds (5, 6). It was soon determined that PCBs adsorbed to soil particulates were eroding from a beach ravine into the marine environment and accumulating in the nearshore sediments. The presence of such compounds in the marine environment is of particular concern due to their tendency to bioaccumulate and biomagnify in the food chain (7–12).

From 1997 to 1999, in conjunction with the excavation and removal of the PCB-contaminated soil from the terrestrial environment at Saglek Bay (13), an extended study was undertaken to assess the contamination in both the marine sediments and biota. An effects-based ecological risk assessment (ERA) was completed to identify potential risks to the local biota (5, 6, 14, 15). Average PCB concentrations in the nearshore marine sediments exceeded the Canadian sediment quality guideline level (21.5 ng/g dry weight (dw)) (16) by 41-fold and PCB concentrations in several wildlife species were well-above Arctic background levels (5). Results of the ERA indicated that the survival and reproduction of shorthorn sculpin (*Myoxocephalus scorpius*) and black guillemot (*Cepphus grylle*) nestlings were at risk (6, 14, 15). By relating levels at which there were effects on their reproduction and survival to concentrations of PCBs in the sediments, it was estimated that risks were posed by sediment PCB concentrations greater than 77 ng/g (dw) for black guillemots and 355 ppb (dw) for sculpin (14). Consequently, the most conservative estimate (77 ng/g) was used as a site-specific risk level. Both shorthorn sculpin and black guillemots are associated with bands of shallow water that occur along coastlines where benthic prey species are relatively abundant (17). Because black guillemots most commonly forage in nearshore areas less than 18 m deep, and are very rarely found at depths greater than 40 m (18, 19), a site-specific Ecological Risk Zone (ERZ) was defined in Saglek Bay as an area adjacent to the former contaminated beach that was less than 40 m deep with sediment PCB concentrations greater than 77 ng/g.

In 2006–07, to gain a better understanding of the current extent of PCB contamination in the marine environment and to determine whether the local ecosystem has been recovering since the removal of the chronic terrestrial PCB source, PCB concentrations were once again measured in sediments, shorthorn sculpin, and black guillemots (nestlings and eggs). Recovery of contaminated marine ecosystems from local point sources is most often a long-term process that has generally received less attention than studies of new contaminated sites, or long-range contaminants, in the Arctic. This study represents one of the first evaluations of Arctic marine ecosystem recovery from a localized PCB source and demonstrates the efficiency with which natural recovery can take place once a chronic input is removed.

Experimental Section

Sample Collection. Sediment, shorthorn sculpin, and black guillemots (nestlings and eggs) were collected in Saglek during two separate time periods: (a) in 1998–2000, concurrent with the terrestrial remediation of PCBs, and (b) eight years later in 2006–07. Sampling in 2006–07 followed the methods outlined in 1998–2000 (see Kuzyk et al. (5) for details). All samples collected during both sampling events were taken from the same areas (Figure 1) at the same time of year (July–August). Surface sediment samples ($n = 92$ in 2006; $n = 112$ in 1998) were collected from four zones in Saglek Bay, up to a depth of 165 m (Figure 1). The four zones were defined

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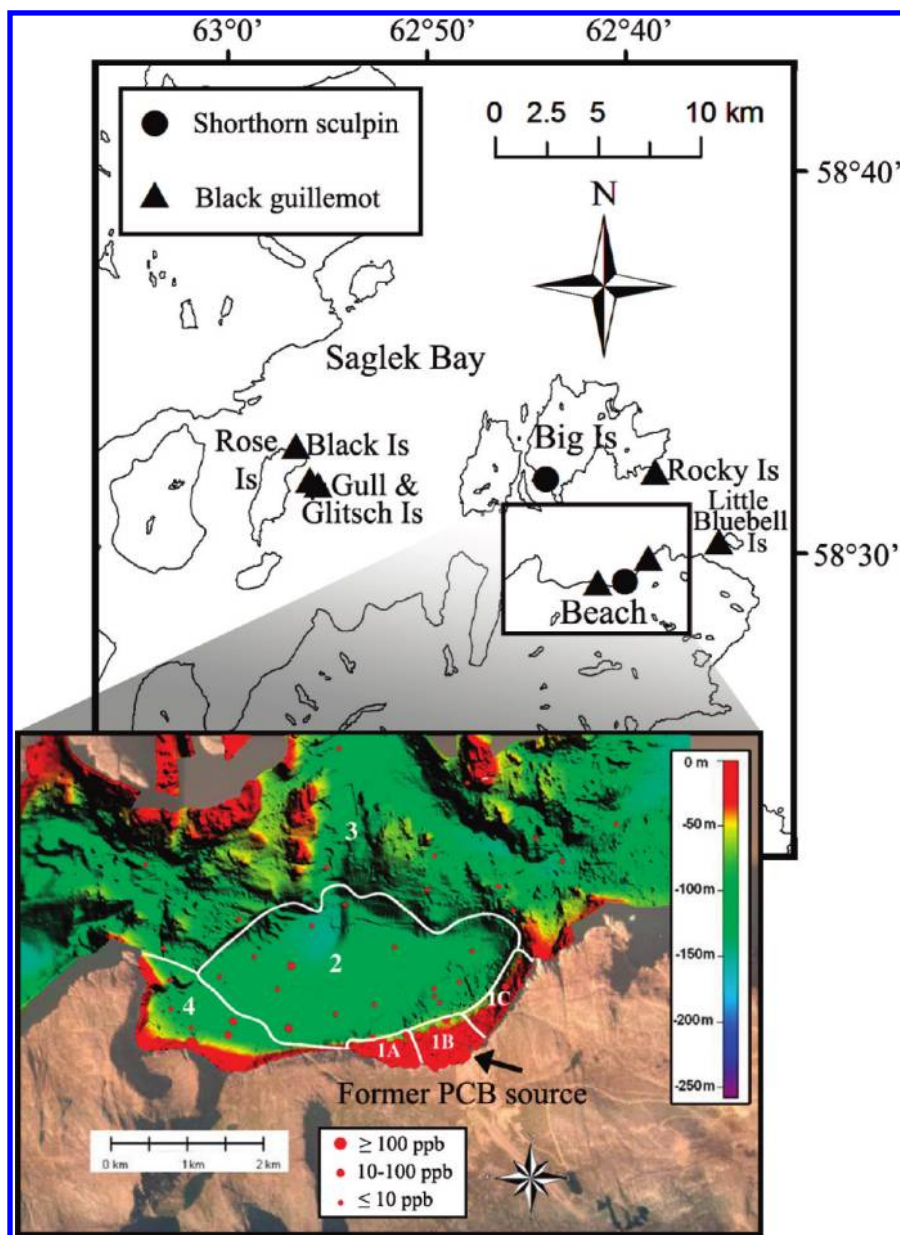


FIGURE 1. Map of Saglek Bay (Labrador, Canada) showing the location of shorthorn sculpin and black guillemot collections. Colored inset map: general bathymetry and the four bathymetric zones in the area of Saglek anchorage, PCB concentrations (ng/g dry weight) in surface sediments, and the location of the former PCB source.

based on bathymetry and PCB distribution to simplify presentation of the data.

Shorthorn sculpin samples were obtained by hook and line from two locations: in the Beach area ($n = 19$ in 2006; $n = 10$ in 1998) adjacent to the former PCB source, and at Big Island ($n = 8$ in 2006; $n = 5$ in 1998) located approximately 6 km to the north (Figure 1). Sculpin were weighed, measured (total length), and immediately dissected to remove the liver for PCB analysis and otoliths for age determination. Gender was determined by gonadal examination and organs were weighed to the nearest 0.1 g on a top-loading balance.

Active black guillemot nests were identified in Saglek Bay at distances of 1–18 km from the contaminated beach (15). The distribution of nests fell into three groups (Figure 1). The first group, hereafter called the “Beach” group, was located on the rocky shoreline within 4 km of the former contaminated beach; the “Island” group was located on two unnamed islands (hereafter called Rocky and Little Bluebell Islands) 5–6 km offshore of the beach; and the “Reference” group was located on Glitsch Island and two unnamed islands

(hereafter called Black and Gull Islands) 16–18 km from the beach (Figure 1). The Reference group serves as a reference for the Saglek Bay area. Short-range transport of PCBs at Saglek resulted in a halo of contamination that was up to 50 km (20); the Reference group is well within the 50 km halo. Nests were visited every 2–3 days, weather permitting, to monitor hatching success and to establish hatch dates. Body size measurements (combined head–bill length) were taken to determine the age of the chicks (Burgess, unpublished data). Chicks were collected for PCB tissue analyses when they were between the ages of 21 and 36 days, with one exception (17 days). A single nestling was removed from each nest; body size and mass were measured. Blood samples were collected for a separate study. These nestlings were then decapitated using a guillotine, allowed to bleed out, and necropsied ($n = 35$ in 2007; $n = 36$ from 1999–2000). For detailed methods related to nestling collection, aging, and necropsy see Kuzyk et al. (15). Livers were dissected from chicks and frozen in acetone/hexane-rinsed glass jars. Black guillemot eggs ($n = 9$ in 2007; $n = 24$ from 1998–99)

TABLE 1. PCB Concentrations [ng/g (Range, 95% UCL)] in Sediment (Dry Weight) and Biota (Lipid Weight) from the Saglek Bay Area Collected in 1998 and 2006–07 (Zone 1 (Nearshore) Is the Sum of Zones 1A–C)

matrix	location	year	N	geometric mean (range)
				(ng/g)
sediment ^a	Zone 1 (nearshore)	2006	51	100 (2.0–800, 143) ^e
		1998	61	1120 (68–62000, 1542)
	A (west)	2006	15	142 (7.0–800, 261)
		1998	3	576 (181–2040)
	B (beach)	2006	30	139 (10–700, 202) ^e
		1998	54	1380 (100–62000, 1901)
	C (east)	2006	6	7.6 (2.0–50, 20) ^e
		1998	4	113 (68–190)
	Zone 2 (deep basin)	2006	16	4.3 (2.0–15, 5.9) ^e
		1998	12	57 (18–183, 84)
shorthorn sculpin ^b (liver)	beach	2006	19	12500 (44–155000, 368129) ^e
		1998	10	242000 (45400–489000, 29242)
	big island	2006	8	450 (256–2270, 735) ^e
		1998	5	1500 (1270–1930, 1866)
	beach west	2007	4	7320 (5130–10400)
		1998	6	7730 (712–80000, 40644)
	beach east	2007	7	562 (291–2640, 991) ^e
		1998	5	11000 (2133–47900, 36898)
black guillemot nestling ^c (liver)	islands	2007	10	165 (69–603, 250) ^e
		1998	10	1020 (404–2290, 1419)
	reference	2007	13	85 (48–193, 105) ^e
		1998	15	175 (100–434, 211)
black guillemot egg ^d	beach	2007	5	9400 (3800–28900, 23550) ^e
		1998	7	67700 (3650–360000, 218273)
	islands	2007	3	3070 (1560–5050)
		1998	11	7370 (3280–34900, 10889)
	reference	2007	1	1670
		1998	6	2960 (1680–4720, 4217)

^a Aroclor equivalents (1260). ^b PCBs include the following: 99, 128, 138/163/164, 151, 153, 156, 170/190, 171, 178, 180, 182/187, 183, 194, 195, 196/203, 199. ^c PCBs: 153, 138/163/164, 182/187, 180. ^d PCBs: 89/90/101, 99, 110, 146, 153, 138/163/164, 128, 156, 178, 182/187, 183, 177, 171, 180, 193, 170/190, 199, 196/203, 194. ^e Significant difference between years $p < 0.05$.

were also collected opportunistically from nests found within the three groups and frozen until analysis.

Sediment Analysis. Sediment samples with expected PCB concentrations greater than 500 ppb were analyzed by the Analytical Services Group (ASG) at the Royal Military College, Kingston, ON, while all low-level analyses were conducted by the AXYS Analytical Services Ltd. for Aroclor equivalents (1260). The congener composition of Saglek Bay sediments strongly resembled the composition of Aroclor 1260, the commercial mixture used at the military station, and 30 samples analyzed by both laboratories showed there was minimal difference between techniques (average CV = 0.13) (5). Sediment samples analyzed at AXYS were assayed by capillary gas chromatography/mass spectrometry (GC/MS) using AXYS in-house methods or an HP 5890 Series II Plus GC equipped with a ⁶³Ni electron capture detector (GC/ECD) (see Supporting Information (SI) for a summary of the method). Control samples, duplicates, and blanks were extracted for 10% of the samples. All values were reported as ng/g dry weight. Detection limits for Aroclor 1260 were sample specific. Acceptable detection limits ranged from 0.01 to 100 ng/g. Two spiked sediment recoveries for Aroclor 1260 averaged 93% ± 5.0 and 95% ± 7.7. One congener spike was outside the control limit of ± 30% recovery at 68%. PCBs were below detection limits (100 ng/g) in blanks ($n = 19$) and all analytical duplicates were within 30% ($n = 3$).

Tissue Analysis. Concentrations of 91 PCB congeners and percent lipid were measured in individual sculpin and guillemot livers as well as guillemot egg samples by AXYS Analytical Services Ltd., Sydney, BC. Samples were analyzed by GC/MS using AXYS in-house methods (see SI for a summary of the method). Detection limits for PCB congeners were sample specific and ranged from 0.01 to 1.0 ng/g (ppb). For analysis of shorthorn sculpin and black guillemot samples collected in 2006–07, PCB concentrations represent ΣPCBs, calculated from the sum of 91 congeners. Because samples collected in 1998–99 were analyzed at a lower resolution and therefore have a higher detection limit compared with 2006–07 samples, the sum of 16, 4, and 19 congeners was used to compare PCB levels from the two time periods for sculpin liver, guillemot nestling liver, and guillemot eggs, respectively (see SI for a summary of congeners). These congeners were always above detection limits in both 1998–99 and 2006–07 samples. For 2006–07 spatial data set only, any congeners less than the detection limit were replaced with a random number between the detection limit and zero. Recoveries of 53 and 25 PCB congeners from 4 and 3 samples of spiked reference materials averaged 104 ± 1.6% and 103 ± 1.5, respectively. Procedural blanks ($n = 7$) were below detection for all congeners. Analytical duplicates were within 30% ($n = 6$). Lipid contents were determined gravimetrically, and are expressed as percentages of wet-tissue

TABLE 2. 2006–07 Geometric Means and Ranges [ng/g Lipid Weight and Wet Weight (Range, 95% UCL)] of PCB Concentrations (Sum of 91 Congeners) in Liver Tissues of Shorthorn Sculpin and Black Guillemot Nestling and Egg Concentrations of Black Guillemots from Saglek Bay

species	location	N	geometric mean (range) ^a	
			(ng/g lipid wt)	(ng/g wet wt)
shorthorn sculpin (liver)	beach	19	16300 (62–165000, 37932)	2940 (14–41000, 6871)
	big island	8	529 (314–2680, 873)	91 (37–226, 139)
black guillemot nestling (liver)	beach	11	2670 (555–20000, 4887)	130 (20–1110, 250)
	islands	10	324 (147–1130, 478)	16 (6.0–77, 24)
	reference	13	185 (110–370, 218)	8.0 (4.0–14, 9.8)
black guillemot egg	beach	5	10900 (4650–32600, 26303)	1060 (450–2790, 2317)
	islands	3	3720 (1880–5760)	410 (206–673)
	reference	1	2140	155

^a PCBs include the following: 5/8, 15, 16/32, 17, 18, 19, 20/21/33, 22, 24/27, 25, 26, 28, 31, 40, 41/64/68/71, 42/59, 43/49, 44, 45, 46, 47/48/75, 51, 52/73, 56/60, 61/74, 62/65, 66/80, 70/76, 83/108, 84, 85/120, 86/97, 87/115/116, 89/90/101, 91, 92, 93/95, 99, 105/127, 106/118, 107/109, 110, 114, 123, 128, 129, 130, 131/142, 132/168, 134/143, 135/144, 136, 137, 138/163/164, 139/149, 141, 146, 151, 153, 156, 157, 158/160, 159, 170/190, 171, 172/192, 174/181, 175, 176, 177, 178, 179, 180, 182/187, 183, 185, 189, 191, 193, 194, 195, 196/203, 197, 198, 199, 201, 205, 206, 207, 208, and 209.

mass. For biological samples, ΣPCB concentrations were expressed on a lipid-weight (lw) basis (Table 1). Table 2 describes the 2006–07 ΣPCB concentrations on both a lw and wet-weight (ww) basis.

Data Analysis. All statistical analyses were performed using SPSS Version 16 for Windows (SPSS, Chicago, IL). PCB data were log transformed to meet the normality assumptions for parametric analyses. Geometric means and ranges were calculated and tissue concentrations were converted to a lipid-weight (lw) basis by dividing wet-weight (ww) values by the proportion of lipid in the samples. For biological samples, ΣPCB concentrations are expressed on a lw basis except where stated otherwise. A one-way ANOVA was used to compare the mean of PCB concentrations among locations within years, by gender within locations, and differences between years. A Tukey's posthoc test was performed to determine differences among locations. Sex ratios and length of sculpin were compared using a Chi-square test. Linear regression analysis was used to determine relationships between PCB tissue concentrations and PCB sediment concentrations and between PCB liver tissue concentrations and age in female or male sculpin. PCB values have been rounded to three significant figures.

Results and Discussion

Surface Sediment PCBs: Spatial and Temporal Trends. Sediment PCB concentrations from zones 1 (A, B, C) to 4 of Saglek Bay are reported in Table 1 with corresponding locations of the zones shown in Figure 1 (colored inset map). In 2006, PCB concentrations in 92 sediment samples ranged from 1 to 800 ng/g (dw) and decreased with increasing distance from the former source of contamination at the beach (Figure 1). PCB concentrations in the surface sediments of zone 1 were significantly higher ($p < 0.05$) than concentrations in sediments from zones 2, 3, and 4. No significant differences were found among zones 2, 3, and 4. PCB concentrations in surface sediments of zone 1A (≤ 500 m west of the beach) and 1B (≤ 500 m north of the beach) were significantly higher ($p < 0.05$) than concentrations detected in sediments of zone 1C (≤ 500 m east of the beach). The maximum PCB concentration (800 ng/g dw) was measured in sediment from a small sand patch approximately 500 m west (zone 1A) of the formerly contaminated beach at a water depth of 3.4 m. PCB concentrations above 100 ng/g (dw) were within 300 m north and 4.0 km west of the former contaminated beach area (Figure 1). Sediment PCB concentrations outside this area quickly fell below 10 ng/g (dw) (Figure 1).

The spatial pattern of the 2006 PCB concentrations in sediment is consistent with results from the 1998–99

investigations (5, 14), with the highest concentrations being reported north and west of the former source. The 2006 surface sediment PCB concentrations in zones 1, 1B, 1C, 2, and 3 decreased significantly ($p < 0.05$) from 1998–99 concentrations (Table 1). The 2006 nearshore (zone 1) sediment concentrations were an order of magnitude lower than the 1998–99 sediment concentrations, with average values falling from 1120 ng/g (dw) in 1998–99 to 100 ng/g (dw) in 2006 ($p < 0.05$) (Table 1). Sediment concentrations from zones 1B, 1C, 2, and 3 also decreased significantly from 1998–99 concentrations (Table 1). Since the removal of the land-based source in 1998–99, there appears to be a trend to lower PCB concentrations immediately west (zone 1A) and further west (zone 4) of the beach but the changes are not statistically significant (Figure 2). The decrease of PCB contamination in the nearshore area indicates that sediment PCB concentrations in the ERZ are decreasing quickly and approaching the site-specific threshold established for probable adverse effects in black guillemot nestlings (77 ng/g) (14).

Due to Saglek's geographical location and bathymetric configuration, the nearshore areas (with the highest PCB concentrations) experience the most frequent sediment mobilization from prevailing currents, tides, and storms (21). Sediment PCB concentrations in 2006 confirm Solomon's (21) prediction that decreases in PCB contamination in the shallow area near the former contaminated beach would occur relatively rapidly (in less than 10 years). Solomon (21) also predicted that fine- to coarse-grained sand would be transported from areas of highest PCB concentrations near the contaminated beach primarily to the west, somewhat to the north, and to a much lesser extent to the east (Figure 2). Our results suggest that the finer sediments are being transported toward the western end of the shallow Saglek Bay area and deposited under the influence of oceanic transport processes. In addition, the contaminated sediments that are being deposited to the north in the deep muddy basin appear to be simultaneously diluted and buried with "clean" sediment resulting in an overall decrease to the surface sediment PCB levels in this area (21).

Sculpin PCBs: Spatial and Temporal Trends. Shorthorn sculpin were collected from two locations: the Beach (< 500 m from the former contaminated beach) and Big Island (6 km north of the former contaminated beach). Ages and lengths ranged from 3–11 years (median 5 years, $n = 22$) and 21.1–36.7 cm (mean 26.4 ± 3.6 cm, $n = 27$), respectively. No significant differences ($p > 0.05$) for sex or average sculpin length were found among the two locations or between the two time periods. ΣPCB concentrations in sculpin expressed

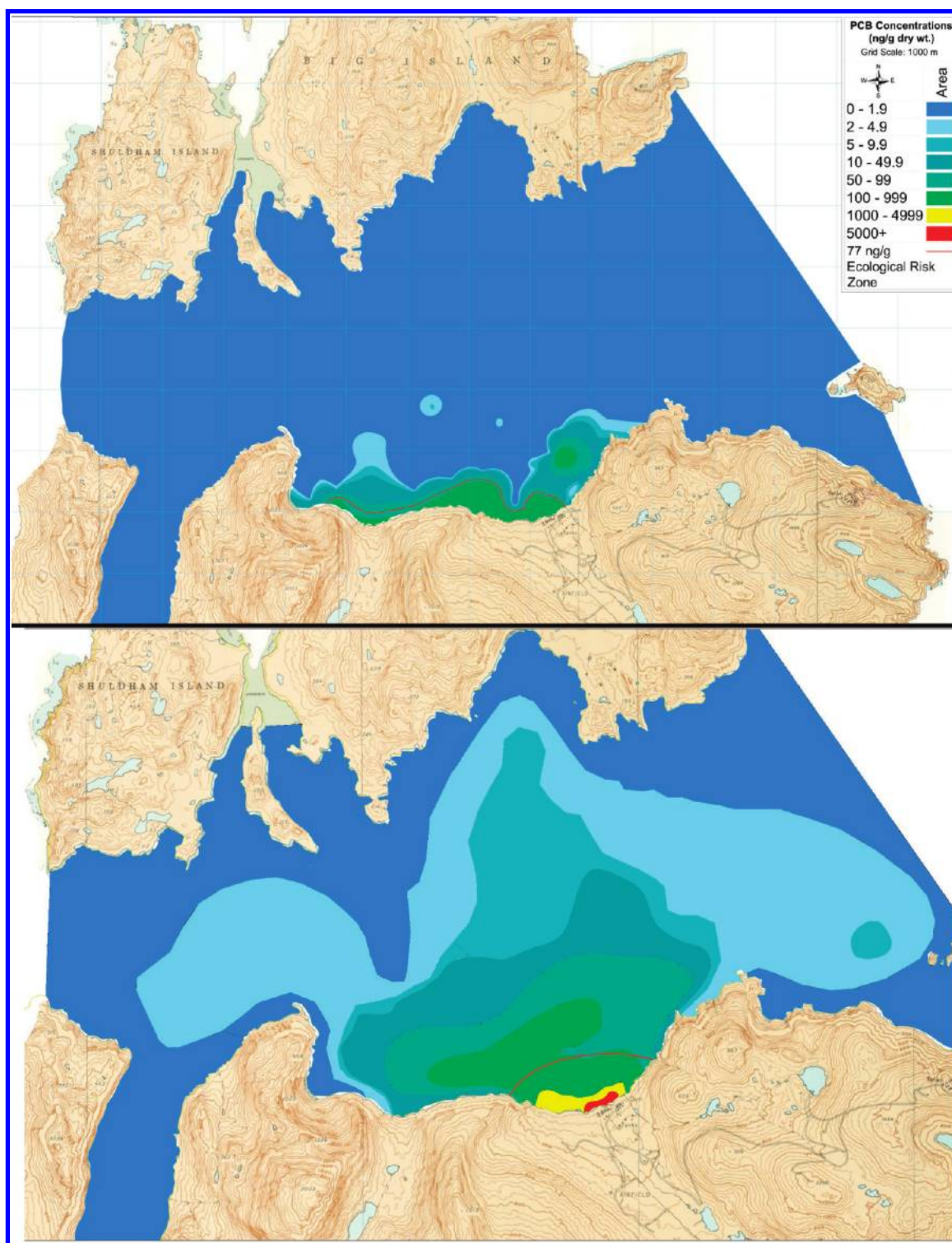


FIGURE 2. PCB concentration (ng/g dry weight) plume maps for surface sediments in Saglek Bay: (top) 2006 PCB concentrations in surface sediments; (bottom) 1998 PCB concentrations in surface sediments.

on a lipid-weight basis showed the same spatial and temporal pattern as wet-weight concentrations.

Shorthorn sculpin Σ PCB concentrations decreased with increasing distance from the former source of PCBs at the beach. The geometric mean concentration of PCBs (sum of all 91 congeners) in sculpin from the beach area was 30 times higher than concentrations from Big Island (Table 2). The elevated sculpin PCB concentrations found in the beach area in 2006 were associated with the highest sediment PCB concentrations, which are found within 500 m of the beach. As well, the relatively lower PCB concentrations found in

2006 sculpin collected from Big Island were associated with the lower sediment concentrations collected from zone 3. These results confirm that the small home range of shorthorn sculpin (22) has resulted in contaminant levels that clearly reflect localized sediment PCB contamination levels, as was found in 1999 (6).

The overall spatial trend in current sculpin PCB levels is similar to the trend observed during 1998–99 (5, 6), where higher concentrations were found closer to the former contaminated beach source. However, average sculpin liver Σ PCB concentrations in both the beach area and Big Island

decreased substantially ($p < 0.05$) between 1998 and 2006 (Table 1). Within the beach area, average sculpin Σ PCB concentrations have decreased approximately 19-fold ($\Delta_{1999-2007} = 230,000$ ng/g lw) while concentrations from sculpin collected at Big Island have decreased 3-fold ($\Delta_{1999-2007} = 1050$ ng/g lw) (Table 1). These reductions over a relatively short period (8 years) reflect the temporal trends that are also being observed in surface sediment PCB concentrations within Saglek Bay.

Despite the strong relationship between PCB levels in the sediment and sculpin, there was a wide range of sculpin Σ PCB concentrations found within each sculpin group in 2006. Sculpin from the Beach group, for example, had a range of 165,000 ng/g (lw) while those from Big Island had a range of 2370 ng/g (lw) (Table 2). These results suggest that other factors, such as individual home range size and foraging behavior and gender and age differences, may be playing a role in exposing sculpin to different levels of PCBs. Furthermore, PCB reductions occurred across all ages and may also be related to the species physiology. Other species of the same genus (*Myoxocephalus*) appear to have a unique PCB metabolic pathway that results in a reduction of as much as 10% of their PCB burden (23).

Sculpin PCBs: Gender and Age Differences. Sculpin livers collected from the Beach group showed a weak tendency for increasing Σ PCB concentrations with age in males ($r^2 = 0.43$, $n = 9$, $p = 0.055$). No significant relationship was found among Σ PCB concentrations with age in females. Relationships between Σ PCB concentrations and sculpin age were also reported for the 1998–99 data (5). In the Beach group, Kuzyk et al. (5) found that there was a weak tendency for increasing Σ PCB concentrations with age in males and decreasing Σ PCB concentrations with age in females. Relationships between Σ PCB concentrations and age have not been reported previously for shorthorn sculpin from other areas; however, this result may be attributed to differences in gender seasonal distributions. Female sculpin generally lay their eggs in November–December in very shallow water (6–11 m) (24). Following spawning, females and immature males move into deep water while mature males remain in shallow water to guard the egg masses (24, 25). Mature male sculpin remaining in the shallow waters at Saglek may explain the relationship of increasing Σ PCB concentrations with age in males. As well, in general, as fish age, PCB levels increase. The lack of relationship found between Σ PCB concentrations and age in females may be due to the offloading of PCBs to their eggs. Some female fish species have been shown to mobilize PCBs from their body into their eggs (26).

Black Guillemot PCBs: Spatial and Temporal Trends. In 2007, PCB analyses were conducted on 11 black guillemot nestling livers from the Beach group, 10 from the Islands group, and 13 from the Reference group (Table 2). Σ PCB concentrations in the 34 livers ranged from 110 ng/g (lw) in the Reference group to 20,000 ng/g (lw) in the Beach group (Table 2). Σ PCB concentrations in the guillemots expressed on a lipid-weight basis showed the same spatial and temporal pattern as wet-weight concentrations.

Guillemot nestling livers collected from the Beach group had an average Σ PCB concentration 8 times greater than guillemot nestlings collected from the Island group and 14 times greater than guillemot nestlings collected from the Reference group (Table 2); both differences were significant ($p < 0.05$). The Σ PCB concentrations in the Island and Reference group were not statistically different. The average guillemot nestling liver Σ PCB concentrations from the Beach group (east and west combined) showed a 6-fold decrease ($\Delta_{1999-2007} = 58,300$ ng/g lw) between 1999 and 2007. When the Beach group was separated into Beach east and Beach west, average nestling Σ PCB concentrations in the Beach west group showed no significant changes between years

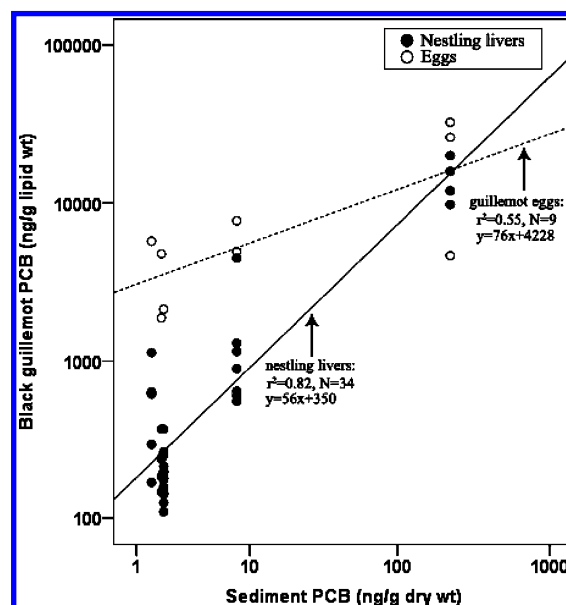


FIGURE 3. Relationship between 2006 surface sediment PCB concentrations (ng/g dry weight) and PCB concentrations in guillemot nestling livers and eggs (ng/g lipid weight).

(Table 1). Average guillemot nestling Σ PCB concentrations in the Beach east, Island, and Reference groups decreased ($p < 0.05$) 20-, 6-, and 2-fold, respectively, from 1999 to 2007 (Table 1). This result is consistent with PCB surface sediment spatial trends indicating that contaminated sediments appear to be migrating mostly to the west and suggests that guillemot nestlings closely reflect sediment PCB conditions.

Σ PCB concentrations in 9 guillemot eggs ranged from 1880 ng/g (lw) in the Islands group to 32,600 ng/g (lw) in the Beach group (Table 2). Eggs from the Beach group had a geometric mean Σ PCB concentration that was 5 times higher than that for eggs collected from the Reference group (Table 2). Σ PCB concentrations in black guillemot eggs from the Beach group decreased ($p < 0.05$) 7-fold from 1999 to 2007 (Table 1). No statistically significant changes from 1999 to 2007 were found in guillemot egg Σ PCB concentrations for the Island and Reference groups (Table 1).

Biotic and Sediment PCB Concentrations: A Strong Relationship. The significant decrease in guillemot nestling and egg PCB concentrations in the Beach group over time is consistent with the pattern of decreasing sediment concentrations observed throughout the nearshore area of the beach (Table 2). Adult guillemots generally forage close to their nesting sites with reported foraging ranges of 0.4–4.0 km (27). Adult guillemot diets primarily consist of benthic fish and invertebrates in waters up to 40 m deep (18, 27) and the diets of nestlings are predominantly fish (27). To determine how well the PCB levels measured in the biota reflect sediment PCB concentrations, an average value derived from sediment concentrations collected within a 500 m radius of each nest location was compared to guillemot liver and egg concentrations (Figure 3). A 500 m radius corresponds to the low end of the range of foraging distances reported in previous studies (27), but was found appropriate by ESG (14) and by observations of foraging guillemots at Saglek. The PCB concentrations in the guillemot livers and eggs correspond closely (r^2 of 0.82 and 0.55 respectively) to the sediment concentrations collected within a 500 m radius of the nests (Figure 3). For any given sediment PCB concentration, eggs typically have higher PCB concentrations than do nestling livers. This pattern is consistent with results reported for various species of birds (28–30). It is believed that the difference can be attributed to the rapid growth of young birds, which has a dilution effect on contaminant

concentrations (31). The strong relationship found for the nestling livers indicates that the guillemot nestlings are closely reflecting PCB conditions in the sediment and suggests that black guillemot nestlings are an excellent indicator species and tool to monitor PCB uptake in local benthic food webs.

Ecosystem Recovery. In 1998–99, studies found evidence of adverse biological effects in sculpin and guillemot nestlings associated with levels of exposure typically found in areas with sediment concentrations greater than 355 and 77 ng/g (dw), respectively (6, 14, 15). The previously defined Ecological Risk Zone (ERZ) was based on the conservative concentration of 77 ng/g (dw). Current sediment concentrations are rapidly approaching this threshold. PCB concentrations in the sediment and biotic indicators (shorthorn sculpin as well as black guillemot nestlings and eggs) continue to decrease rapidly over time. Black guillemot nestling PCB concentrations are now below concentrations (1000–2750 ng/g ww) previously associated with risks of impaired reproduction and survival (14, 15). The guillemot nestling liver PCB exposures were established upon the completion of the Saglek effects-based ecological risk assessment (14, 15). Sculpin liver PCB concentrations, however, are still above the lower range of thresholds thought to impair fish reproduction (50 ng/g ww) (32). PCB residues and associated risks are still present, however, as the majority of the sediment samples collected in the ERZ exceed the previously determined site-specific target level above which ecological effects are probable (77 ng/g dw).

Sediment PCB concentrations in the ERZ have decreased significantly, although the ERZ is undergoing a slight westward migration along the immediate nearshore area of the beach (Figure 2). The 2006 sediment results support a previous qualitative prediction that natural processes would result in rapid decreases (over 5–10 years) in PCB concentrations in the shallow beach area near the former source of PCBs (21). Over a period of 8 years, a 2–14 fold reduction has been measured in Saglek Bay surface sediments. These decreases are comparable with the recovery of PCB contaminated sediments at Lake Hartwell, South Carolina (Superfund Site), where a 10–50 fold reduction was measured over a 25 yr period (33).

The decreases observed in the two indicator species and the strong association found between sediment and biota suggest that the indicator species closely reflect the present-day PCB contamination at Saglek Bay and that natural ecosystem recovery is occurring. Biotic recovery trends observed in this study differ from the biotic trends reported at Lake Hartwell, where largemouth and hybrid bass have not measurably responded to decreases in surface sediment PCB concentrations (33). The time lag between surface sediment PCB concentrations and the response of indicator species at Lake Hartwell appears to be far greater than that observed at Saglek Bay. A possible explanation for this is the differing life history traits among the indicator species in the two studies. Largemouth bass are piscivorous, feeding mostly on fish at higher trophic levels (34), whereas shorthorn sculpin feed lower in the food chain, with a diet consisting of mainly benthic invertebrates (22).

In many studies of local sources of contamination, natural variability can confound and dilute statistical trends. By using guillemot chicks of known age and sex, with no differences in body weight, we were able to control for much of the natural variability, allowing a sound comparison of PCB concentrations between years and location. The results of our study demonstrate the utility of using indicator species, especially black guillemots, to study the recovery of marine ecosystems in northern coastal environments. This study illustrates how quickly both the physical and biotic PCB concentrations of a highly energetic coastal ecosystem can decrease once a point source is removed. This is an important

first part of ecosystem recovery, and companion studies (manuscript in preparation) have shown that the decline in the PCB concentrations are associated with a decline in biological effects.

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Supporting Information Available

Summary of analytical methods for sediment and biotic analyses; summary of congeners for spatial data set (2006–07) and temporal data set (1998–99 and 2006–07) for biota. This information is available free of charge via the Internet at <http://pubs.acs.org/>.

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